

Question 1:

From LLAMA 2 we know that $N = K$. Therefore:

$$\text{MLP Params} = 3DF \quad \text{per layer}$$

$$\text{Attention Params} = 4DNH \quad \text{per layer}$$

$$\text{Embeddings} = 2DV \quad \text{Overall.}$$

In total we have $(3DF + 4DNH) \cdot L + 2DV$ parameters. Let $L = 40$, $N \cdot H = D$, $D = 5120$, $F = 13824$, $V = 32000$. Hence:

$$\begin{aligned} & (3 \cdot 5120 \cdot 13824 + 4(5120)^2) \cdot 40 + 2 \cdot 5120 \cdot 32000 \\ &= 13015449600 \approx 13 \cdot 10^9 = 13 \text{ Billion.} \end{aligned}$$

Question 2:

$$\text{Model Params} = 2 \cdot 13 \cdot 10^9 = 26 \cdot 10^9 \text{ Bytes} = 26 \text{ GB.}$$

$$\text{Optimizer State} = 4 \cdot 13e9 + 4 \cdot 13e9 = 8 \cdot 13e9 = 104 \text{ GB.}$$

$$\begin{aligned} \text{Activation Checkpointing} &= L \cdot (2BTF + 2BTF + 2BD) \\ &= 40 \cdot (4BTF + 2BD) = 40 \cdot (BT) \cdot (4F + 2D) \\ &= 40 \cdot (16e6) \cdot (4 \cdot 13824 + 2 \cdot 5120) \\ &\approx 42 \text{ TB} \end{aligned}$$

Question 3:

let:

$$\text{HBM vSp} = 96 \text{ GB}$$

$$\text{bf16 FLOPs/s vSp} = 4.54e14$$

$$\text{WIC1 vSp} = 1.8e11$$

3.1

We cannot use data parallelism since a lower bound in memory per chip we need is 130 GB. (In fact we need more if we consider the activation checkpoints).

3.2

Assume FSOP shards the whole architecture as follows. For the attention params:

$$W_q \in \mathbb{R}^{D_x, N, H}$$

$$W_k \in \mathbb{R}^{D_x, N, H}$$

$$W_v \in \mathbb{R}^{D_x, N, H}$$

$$W_o \in \mathbb{R}^{N, H, D_x}$$

and for the MLP part as usual:

$$W_{in1} \in \mathbb{R}^{D_x, F}$$

$$W_{in2} \in \mathbb{R}^{D_x, F}$$

$$W_{out} \in \mathbb{R}^{F, D_x}$$

and for input output embeddings as:

Embedded $[V, D \times S]$

Woutembled $[D \times, V]$

Therefore the total params per device is

$$\left(3 \frac{D}{|X|} F + 4 \frac{D}{|X|} H N \right) \cdot L + 2 \frac{D}{|X|} \cdot V =$$

$$= \frac{1}{|X|} \cdot \left((3DF + 4DHN) \cdot L + 2 \cdot DV \right)$$

$$= \frac{\text{Model Params}}{|X|}$$

For LLaMA 2 13B therefore:

$$\text{Device Params} = \frac{13 \cdot 10^9}{16^3} \approx 3.2 \cdot 10^6 = 3.2 \text{ Million}$$

Assuming optimizer states also sharded, we get that each device needs at least:

$$10 \cdot (3.2 \cdot 10^6) = 32 \cdot 10^6 = 32 \text{ MB}$$

If we also take into account the

activation checkpoints, which are of shapes:

$$[B_x, T, F]$$

$$[B_x, T, F]$$

$$[B_x, T, D]$$

we get:

$$\begin{aligned} \text{Checkpoints Per Device} &= L \cdot \left(2 \frac{B}{|X|} \cdot TD + 4 \frac{B}{|X|} TF \right) \text{ Bytes} \\ &= \frac{1}{|X|} \cdot \left(L \cdot (BT) \cdot (4F + 2D) \right) \text{ Bytes} \end{aligned}$$

let $BT = 3 \cdot 10^6$, then:

$$\frac{1}{16^3} \cdot \left(40 \cdot (3 \cdot 10^6) \cdot (4 \cdot 13824 + 2 \cdot 5120) \right)$$

$$\approx 1.96 \text{ GB.}$$

So memory used per device is $32 \text{ MB} + 1.96 \text{ GB}$.

We could use FSDP since with FSDP sharding everything fits in memory

but we would be communication bound
since we are compute bound iff:

$$\frac{B}{|X|} > \frac{2550}{M_x} \Leftrightarrow \frac{M_x \cdot B}{2550} > |X|$$

let $B = 3 \cdot 10^6$ and $M_x = 3$, therefore:

$$\frac{4 \cdot 10^6}{2550} \simeq 3530 > |X|$$

and in our slice $|X| = 4096$.

3.2

Assume we shard the attention matrices as:

$$W_q [D_x, N_y, H]$$

$$W_k [D_x, N_y, H]$$

$$W_v [D_x, N_y, H]$$

$$W_o [N_y, H, D_x]$$

and for the MLP params as:

$$W_{in1} [D_x, F_y]$$

$$W_{in} \in [D_x, F_y]$$

$$W_{out} \in [F_x, D_x]$$

and for input and output embeddings as:

$$V_{embed} \in [V_y, D_x]$$

$$V_{out_embed} \in [D_x, V_y]$$

Therefore total params per device is:

$$\left(3 \frac{D \cdot F}{|X| \cdot |Y|} + 4 \frac{D \cdot H \cdot N}{|X| \cdot |Y|} \right) \cdot L + 2 \frac{D \cdot V}{|X| \cdot |Y|} =$$

$$= \frac{1}{|X| \cdot |Y|} \cdot \left(L \cdot (3DF + 4DHN) + 2 \cdot DV \right)$$

$$= \frac{\text{Model Params}}{|X| \cdot |Y|}$$

For LLaMA2 13B we get:

$$\text{Params Per Device} = \frac{13 \cdot 10^9}{(16)^3} \approx 3.2 \text{ Million}$$

if we take into account activation states then we get:

$$10 \cdot (3.2 \cdot 10^6) = 32 \text{ MB}$$

Activation check points will be sparded with shapes:

$$[B_x, T, F_y]$$

$$[B_x, T, F_y]$$

$$[B_x, T, D_y]$$

so we get:

$$\begin{aligned} \text{Checkpoints Per Device} &= L \cdot \left(2 \frac{B}{|X|} \cdot \frac{T D}{|Y|} + 4 \frac{B}{|X|} \frac{T F}{|Y|} \right) \text{ Bytes} \\ &= \frac{1}{|X||Y|} \cdot \left(L \cdot (B T) \cdot (4 F + 2 D) \right) \text{ Bytes} \end{aligned}$$

let $B T = 3 \cdot 10^6$, then :

$$\frac{1}{16^3} \cdot \left(40 \cdot (3 \cdot 10^6) \cdot (4 \cdot 13824 + 2 \cdot 5120) \right)$$

$$\approx 1.4 \text{ GB.}$$

Therefore for memory used per device we get ≈ 1.9 GB.

If we use FSDP and TP to minimize communications we need to set

$$X_{opt} = \sqrt{\frac{B}{F} \cdot \frac{M_x}{M_y} \cdot N}$$

let $N = 16^3$, $M_x = 2$, $M_y = 1$, $B = 3 \cdot 10^6$
and $F = 13824$. Therefore:

$$X_{opt} = \sqrt{\frac{3 \cdot 10^6}{13824} \cdot 2 \cdot 16^3}$$

$$\approx 1773.3 \approx 1773$$

$$\text{Since } N = X_{opt} \cdot Y \Rightarrow \frac{16^3}{1773} = Y = 3.07$$

to get integer values therefore we choose:

$$X_{opt} = 1024 \quad \text{and} \quad Y = 4$$

which is closer to our optimum.

Having chosen this, we will be compute bound iff:

$$\frac{B}{N} > \frac{\alpha^2}{M_x M_y F}$$

where $\alpha = \frac{C}{W_{II}} = 2550$ for v5p.

let $F = 13824$, $N = 16^3$, $M_x = 2$, $M_y = 1$
and $B = 3 \cdot 10^6$:

$$\frac{B}{N} = \frac{3 \cdot 10^6}{16^3} \approx 733$$

$$\frac{\alpha^2}{M_x M_y \cdot F} = \frac{(2550)^2}{2 \cdot 13824} = 235$$

therefore we are compute bound.

Assuming we are compute bound, using only roof line FLOPs estimates and ignoring attention we get that:

$$\text{Time Per Step} = \frac{3 \times 10^6 \cdot 13 \times 10^9 \cdot 6}{16^3 \cdot (0.4 \cdot 4.59 \times 10^{14})}$$

$$= 0.312 \text{ s}$$

$$= 312 \text{ ms}.$$